

Multiple Choice

1. **Answer:** B

Options: A) Increase the amount of training data.; B) Improve the optimisation algorithm being used for error minimisation.; C) Decrease the model complexity.; D) Reduce the noise in the training data.

Note: Correct option: B - Improve the optimisation algorithm being used for error minimisation.

2. **Answer:** D

Options: A) It can not be applied to non-differentiable functions.; B) It can not be applied to non-continuous functions.; C) It is hard to implement.; D) It runs reasonably slow for multiple linear regression.

Note: Correct option: D - It runs reasonably slow for multiple linear regression.

3. **Answer:** D

Options: A) It relates inputs to outputs.; B) It is used for prediction.; C) It may be used for interpretation.; D) It discovers causal relationships

Note: Correct option: D - It discovers causal relationships

4. **Answer:** A

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Correct option: A - True, True

5. **Answer:** B

Options: A) The use of sampling with replacement as the sampling technique; B) The use of weak classifiers; C) The use of classification algorithms which are not prone to overfitting; D) The practice of validation performed on every classifier trained

Note: Correct option: B - The use of weak classifiers

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** `def __init__(self, data): | def max_height(node): | return 0 ; | return left_height+1 | return right_height+1`
Note: Students should provide a detailed explanation referencing algorithm steps.
12. **Answer:** `def count_elim(num): | return count_elim`
Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key